

SAIF-ALDAIN AQEL
FINAL PROJECT



M Ű E G Y E T E M 1 7 8 2

BUDAPEST UNIVERSITY OF TECHNOLOGY AND ECONOMICS

FACULTY OF MECHANICAL ENGINEERING

DEPARTMENT OF BUILDING SERVICE ENGINEERING AND PROCESS
ENGINEERING



CAMILLE MARIE CARDOZE

Final Project

Regeneration of a Pharmaceutical Waste Solvent by Batch
Extractive Distillation

Supervisor:

Dr. László Hégely, Associate Professor

Budapest, 2026

DECLARATIONS

Declaration about the acceptability of the thesis

This thesis fulfills every formal and content requirements of the regulation of the Budapest University of Technology and Economics, moreover it fulfills the assignment of the final project. This thesis is suitable for a review and an open defence.

Budapest, DATE

Supervisor's signature

Supervisor's name

Declaration about the independent work

I, STUDENT'S NAME (NEPTUN CODE), hereby declare that the Thesis submitted for assessment and defence, exclusively contains the results of my own work assisted by my supervisor. Further to it, it is also stated that all other results taken from the technical literature or other sources are clearly identified and referred to according to copyright (footnotes/references are chapter and verse, and placed appropriately).

I accept that the scientific results presented in my Thesis can be utilised by the Department of the supervisor for further research or teaching purposes.

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PREAMBLE

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Acknowledgements should be included here.

Budapest, 20...

Student's Name

NOMENCLATURE

This table summarizes the reoccurring physical quantities with their names and units. Please use the same notations as employed by textbooks and scientific journals wherever possible. If a quantity is scarcely used, their description should be included at their first appearance.

The table should be in alphabetic order, starting with lowercase letters. If a notation corresponds to multiple quantities, they should be described in individual lines. The values must be included in the case of constants (see the example below). The unit of dimensionless quantities is 1 (again, see below). SI units should be used, and avoid using both metric and imperial units. This paragraph should be deleted from the final document.

Latin symbols

Notation	Name, notes, value	Unit
g	gravitational acceleration (9.81)	m/s ²
p	pressure	bar
c_D	drag coefficient	1
s	specific entropy	J/(kg·K)

Greek symbols

Notation	Name, notes, value	Unit

Subscripts and superscripts

Notation	Name, notes, value
i	general index (integer)
nom	nominal value
opt	optimal value

1. INTRODUCTION

1.1. Goals

Intendation should be avoided in the first paragraphs, and it should be 0.4 cm for the later ones.

Equations must be written using the following format:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1.1)$$

Equations should be numbered and referred to by their numbers in the text. The numbering can be continuous (e.g. 1, 2, 3...) or might restart in each chapter (e.g. 1.1, 1.2, 1.3, 2.1...).

Harvard citation style should be used and all references must appear in the text in brackets with the name of the authors' and the year of the publication. Both curved () and square [] brackets are permitted. In the case of more than two authors, the „et al.” notation must be used. Examples: (Smith, 2012), (Smith and Johnson, 2009), (Smith et al., 2007). In the case of multiple publications in the same year by the same author, use letters in alphabetical order to distinguish them, for example (Smith, 2012a) and (Smith, 2012b). If the name of the author is also mentioned in the text, then only the year should be in the brackets, for example: „After extensive measurements, Smith (2012) was proven wrong”. The references should be listed at the end of the document in alphabetical order by the first author's surname (see 6.).

1.2. Overview

What is in this document and in what order.

1.2.1. PRINCIPLES AND METHODS

1.2.1.1. First method

1.2.1.2. Second method

1.2.2. ASSUMPTIONS

2. LITERATURE REVIEW/PREVIOUS STUDIES

2.1. For the first part

2.2. For the second part

3. FIRST TOPIC

3.1. *First subtopic*

3.2. *Second subtopic*

3.3. *Nth subtopic*

4. SECOND TOPIC

4.1. *First subtopic*

There should be at most two topics.

4.2. *Second subtopic*

4.3. *Nth subtopic*

5. CONCLUSIONS/OVERVIEW

5.1. Results

The overview of the results should be at most three pages. Anything regarding the work of the student (i.e. you) should be in first person.

5.2. Suggestions/Conclusions/Summary of Experience

Summary of the conclusions regarding the whole work. Suggestions regarding future work on the topic.

6. REFERENCES

1. AUTHORS' NAME (1999): *Title*. Publisher, City of the publisher. Other IDs (e.g. doi, url)
2. AAAA
3. BBBB
4. CCCC

7. SUMMARY

Brief summary (at most one and a half pages, but at least 2/3rd of a page) of the work in Hungarian. *Only needed if the author is Hungarian.*

Keywords: *keyword1, keyword2, keyword3*